

# Лабораторная работа №6

Установка и настройка СУБД MariaDB

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24 ноября 2025

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## Цель работы

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Приобрести практические навыки по установке, настройке и администрированию СУБД **MariaDB** на примере виртуальной машины *server*.

## Установка MariaDB

---

```
Installed:
 mariadb-3:10.11.11-1.el10.x86_64
 mariadb-client-utils-3:10.11.11-1.el10.x86_64
 mariadb-errmsg-3:10.11.11-1.el10.noarch
 mariadb-server-3:10.11.11-1.el10.x86_64
 mysql-selinux-1.0.14-1.el10_0.noarch
 perl-Sys-Hostname-1.25-512.2.el10_0.x86_64
 mariadb-backup-3:10.11.11-1.el10.x86_64
 mariadb-common-3:10.11.11-1.el10.noarch
 mariadb-gssapi-server-3:10.11.11-1.el10.x86_64
 mariadb-server-utils-3:10.11.11-1.el10.x86_64
 perl-DBD-MariaDB-1.23-10.el10.x86_64

Complete!
[root@server.vabazlov.net ~]# systemctl start mariadb
[root@server.vabazlov.net ~]# systemctl enable mariadb
Created symlink '/etc/systemd/system/mysql.service' → '/usr/lib/systemd/system/mariadb.service'.
Created symlink '/etc/systemd/system/mysqld.service' → '/usr/lib/systemd/system/mariadb.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/mariadb.service' → '/usr/lib/systemd/system/mariadb.service'.
[root@server.vabazlov.net ~]# ss -tulpen | grep mysql
[root@server.vabazlov.net ~]# ss -tulpen | grep maria
tcp LISTEN 0      80          0.0.0.0:3306  0.0.0.0:*    users:(("mariadb",pid=14333,fd=20))
                                     uid:27 ino:56850 sk:18 cgroup:/system.slice
/mariadb.service <->
tcp LISTEN 0      80          [::]:3306    [::]:*      users:(("mariadb",pid=14333,fd=21))
                                     uid:27 ino:56851 sk:24 cgroup:/system.slice
/mariadb.service v6only:1 <->
[root@server.vabazlov.net ~]#
```

Рис. 1: Запуск службы MariaDB

```
Disallow root login remotely? [Y/n]
```

```
... Success!
```

By default, MariaDB comes with a database named 'test' that anyone can access. This is also intended only for testing, and should be removed before moving into a production environment.

```
Remove test database and access to it? [Y/n]
```

```
- Dropping test database...
```

```
... Success!
```

```
- Removing privileges on test database...
```

```
... Success!
```

Reloading the privilege tables will ensure that all changes made so far will take effect immediately.

```
Reload privilege tables now? [Y/n]
```

```
... Success!
```

```
Cleaning up...
```

All done! If you've completed all of the above steps, your MariaDB installation should now be secure.

```
Thanks for using MariaDB!
```

```
[root@server.vabazlov.net ~]#
```

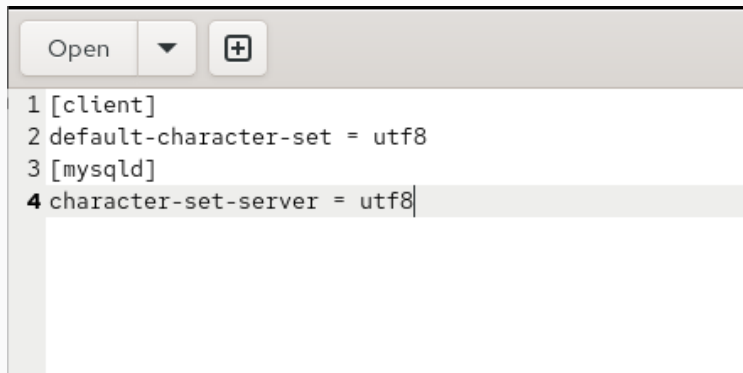
```
MariaDB [(none)]> status
-----
mysql Ver 15.1 Distrib 10.11.11-MariaDB, for Linux (x86_64) using EditLine wrapper

Connection id:          11
Current database:
Current user:           root@localhost
SSL:                    Not in use
Current pager:          stdout
Using outfile:          ''
Using delimiter:        ;
Server:                 MariaDB
Server version:         10.11.11-MariaDB MariaDB Server
Protocol version:       10
Connection:             Localhost via UNIX socket
Server characterset:    latin1
Db characterset:        latin1
Client characterset:    utf8mb3
Conn. characterset:     utf8mb3
UNIX socket:            /var/lib/mysql/mysql.sock
Uptime:                 3 min 24 sec

Threads: 1  Questions: 21  Slow queries: 0  Opens: 20  Open tables: 13  Queries per second avg: 0.102
-----

MariaDB [(none)]> █
```

Рис. 3: Статус до настройки



The image shows a text editor window with a light gray header bar containing three buttons: 'Open', a dropdown arrow, and a '+' icon. The main area of the editor is white and contains the following text:

```
1 [client]
2 default-character-set = utf8
3 [mysqld]
4 character-set-server = utf8|
```

The fourth line is highlighted with a light gray background, and a vertical cursor is positioned at the end of the text on that line.

Рис. 4: Конфигурация utf8.cnf



```
MariaDB [(none)]> status
-----
mysql Ver 15.1 Distrib 10.11.11-MariaDB, for Linux (x86_64) using EditLine wrapper

Connection id:          3
Current database:
Current user:           root@localhost
SSL:                    Not in use
Current pager:          stdout
Using outfile:          ''
Using delimiter:        ;
Server:                 MariaDB
Server version:         10.11.11-MariaDB MariaDB Server
Protocol version:       10
Connection:             Localhost via UNIX socket
Server characterset:    utf8mb3
Db characterset:        utf8mb3
Client characterset:    utf8mb3
Conn. characterset:     utf8mb3
UNIX socket:            /var/lib/mysql/mysql.sock
Uptime:                 8 sec

Threads: 1  Questions: 4  Slow queries: 0  Opens: 17  Open tables: 10  Queries per second avg: 0.500
-----

MariaDB [(none)]> █
```

Рис. 5: Статус после настройки

## Создание базы и таблицы

```
MariaDB [(none)]> CREATE DATABASE addressbook CHARACTER SET utf8 COLLATE utf8_general_ci;  
Query OK, 1 row affected (0.000 sec)
```

```
MariaDB [(none)]> USE addressbook;  
Database changed  
MariaDB [addressbook]> SHOW TABLES;  
Empty set (0.000 sec)
```

```
MariaDB [addressbook]> CREATE TABLE city(name VARCHAR(40), city VARCHAR(40));  
Query OK, 0 rows affected (0.006 sec)
```

```
MariaDB [addressbook]> INSERT INTO city(name, city) VALUES ('Иванов', 'Москва');  
Query OK, 1 row affected (0.002 sec)
```

```
MariaDB [addressbook]> INSERT INTO city(name, city) VALUES ('Петров', 'Сочи');  
Query OK, 1 row affected (0.001 sec)
```

```
MariaDB [addressbook]> INSERT INTO city(name, city) VALUES ('Сидоров', 'Дубна');  
Query OK, 1 row affected (0.001 sec)
```

```
MariaDB [addressbook]> SELECT * FROM city;
```

| name    | city   |
|---------|--------|
| Иванов  | Москва |
| Петров  | Сочи   |
| Сидоров | Дубна  |

```
3 rows in set (0.000 sec)
```

```
MariaDB [addressbook]> █
```

```
MariaDB [addressbook] /
MariaDB [addressbook]> CREATE USER vabazlov@'%' IDENTIFIED BY '123456';
Query OK, 0 rows affected (0.001 sec)

MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO vabazlov@'%';
Query OK, 0 rows affected (0.000 sec)

MariaDB [addressbook]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.001 sec)

MariaDB [addressbook]> DESCRIBE city;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| name  | varchar(40)   | YES  |     | NULL    |       |
| city  | varchar(40)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.001 sec)

MariaDB [addressbook]> quit
Bye
[root@server.vabazlov.net my.cnf.d]#
```

Рис. 7: Структура таблицы city

```

[root@server.vabazlov.net my.cnf.d]# mkdir -p /var/backup
[root@server.vabazlov.net my.cnf.d]# mysqldump -u root -p addressbook > /var/backup/addressbook.sql
Enter password:
[root@server.vabazlov.net my.cnf.d]# mysqldump -u root -p addressbook | gzip > /var/backup/addressbook.sql.gz
Enter password:
[root@server.vabazlov.net my.cnf.d]# mysqldump -u root -p addressbook | gzip > $(date +%Y%m%d.%H%M%S).sql.gz)
-bash: syntax error near unexpected token `)'
[root@server.vabazlov.net my.cnf.d]# mysqldump -u root -p addressbook | gzip > $(date +%Y%m%d.%H%M%S).sql.gz)
Enter password:
[root@server.vabazlov.net my.cnf.d]# ls
auth_gssapi.cnf  enable_encryption.preset  mysql-clients.cnf  provider_lz4.cnf  provider_snappy.cnf  utf8.cnf
client.cnf      mariadb-server.cnf        provider_bzip2.cnf  provider_lzo.cnf  spider.cnf
[root@server.vabazlov.net my.cnf.d]# cd /var/backup/
[root@server.vabazlov.net backup]# ls
addressbook.20251124.072006.sql.gz  addressbook.sql  addressbook.sql.gz
[root@server.vabazlov.net backup]# mysql -u root -p addressbook < addressbook.sql
Enter password:
[root@server.vabazlov.net backup]# zcat addressbook.sql.gz | mysql -u root -p addressbook
Enter password:
[root@server.vabazlov.net backup]# █
```

Рис. 8: Восстановление БД

## Итоги работы

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В ходе выполнения лабораторной работы: - установлена и настроена СУБД MariaDB; - выполнена настройка безопасности и кодировок; - создана БД **addressbook** и пользовательские записи; - освоены операции резервного копирования и восстановления; - подготовлена инфраструктура для автоматизированного деплоя.